

Ensemble Spread Is Not Posterior Uncertainty

A width-invariance test, and a bridge between an open question and an unused answer

Sammuel Pendry (framework and mechanism), with formalization and testing by Claude (Anthropic)

CC BY 4.0 | Version 1.0, July 26, 2026

Summary for the impatient

Two published papers answer each other and neither one knows it.

One asks why the epistemic uncertainty reported by deep ensembles shrinks as networks grow, calls the effect a paradox, and closes by saying the reasons have yet to be found. The other proves that deep ensembles perform empirical Bayes under a prior learned implicitly from the data rather than chosen by the practitioner. Read together, the paradox stops needing a cause.

We supply the measurement that joins them. On a regression task with a deliberate hole in the training support, we compare the ensemble spread that practitioners read as epistemic uncertainty against the exact Gaussian process posterior under each width's own implicit prior, which is the correct Bayesian answer for that architecture. Across a 32x range of width the correct answer holds flat at slope +0.004 per doubling. The ensemble spread falls by a factor of 7.7 over the same range.

The ignorance implied by the model does not change with width. The number practitioners read as ignorance does. The collapse belongs to the estimator.

Three findings follow. No fixed rescaling repairs it, because the distortion depends on both width and region. The ensemble is heading toward a computable asymptote that sits 42 percent below the correct posterior, so it crosses from overstating ignorance to understating it. And the asymptotic understatement runs worse inside the training data than outside it, which we did not expect and traced to a structural gap between two kernels rather than to anything about extrapolation.

This paper carries 4 failures and 3 open tests.

What this paper is

A small empirical result and a bridge. We claim no theory. The mechanism belongs to the empirical Bayes paper, the phenomenon belongs to the uncertainty calibration papers, and the measurement connecting them is what we added.

We name the two papers we hope find each other:

- *On the Calibration of Epistemic Uncertainty: Principles, Paradoxes and Conflictual Loss*, arXiv:2407.12211, which documents the collapse, introduces conflictual deep ensembles as a repair, and states in its conclusion that the exact reasons for the collapse under growing complexity have yet to be found. That sentence is why this paper exists.
- *Deep Ensembles Secretly Perform Empirical Bayes*, arXiv:2501.17917, which shows deep ensembles perform exact Bayesian averaging under a prior learned implicitly from the data.

Rules fixed before the test ran. We set the construction, the width sweep, the seed count, and the comparison quantity in advance, and stated the kill condition first: if the Gaussian process posterior in the hole tracked the ensemble collapse, the artifact reading was wrong and we would report that instead.

This work began inside a different framework, on representation systems under finite description budgets. That framework died, partly to its own data and partly to prior art we found after believing we had something. Its ledger stays with it and does not appear here. Two pieces carry forward: the support-hole construction in A.1, and the search discipline that now precedes every novelty claim we make.

Part A: The result

A.1 The construction

Take a one dimensional regression target, $y = \sin(2.5x) + 0.4x$, with Gaussian noise at 0.05. Draw 2000 training points from $[-3, -0.8] \cup [0.8, 3]$. The interval $[-0.6, 0.6]$ holds no training support. Call it the hole. Inside the hole the data imposes nothing, so whatever a model does there comes from inductive bias rather than evidence. A

second probe region inside the support, [1.0, 2.8], gives the constrained comparison.

Train tanh multilayer perceptrons of depth 3 at widths 8 through 256, eight to twelve seeds each. Measure across-seed standard deviation of the prediction in both regions, normalized by the standard deviation of the training target.

A.2 The two quantities

Ensemble spread. The across-seed standard deviation. Practitioners read this as epistemic uncertainty and use it for out-of-distribution detection, active learning, and safety gating.

Implied posterior uncertainty. For each width, draw networks at initialization and evaluate them on the probe points. Their output covariance is the width- w implicit prior, the NNGP kernel, estimated without training anything. Condition that prior on the training data by exact Gaussian process regression. The resulting posterior standard deviation is the correct Bayesian uncertainty for a model with that architecture and that data. No optimizer enters it and no ensemble enters it.

A.3 The measurement

Width	Prior std, hole	Posterior std, hole	Posterior std, support	Ensemble spread, hole
8	0.2780	0.0151	0.0080	0.7409
16	0.1975	0.0158	0.0080	0.4549
32	0.1637	0.0156	0.0080	0.2647
64	0.1387	0.0153	0.0080	0.3487
128	0.1114	0.0153	0.0080	0.1783
256	0.1041	0.0157	0.0080	0.0955

Log slopes per doubling: prior -0.280 , posterior in hole $+0.004$, posterior in support -0.001 , ensemble spread -0.527 .

The prior does tighten with width, by a factor of 2.7 across the sweep, which matches what anyone would expect from a kernel built by summing w near-independent contributions. The posterior does not tighten at all. Conditioning on 2000 points overwhelms the change in prior scale, and the implied ignorance in the hole lands on the same value at every width. That contrast, a 2.7x prior contraction producing zero posterior movement, is the finding.

The ensemble tracks neither. Its slope sits near the $1/\sqrt{w}$ rate that kernel fluctuation predicts.

A.4 Why no rescaling repairs it

Inside the hole the ensemble overstates the posterior by 49x at width 8 and by 6x at width 256. Inside the training support the overstatement holds near 4x at all widths. A calibration constant fitted on in-distribution data would be wrong out of distribution by a factor that itself depends on width. The distortion has two dimensions, region and capacity, and a scalar cannot absorb it.

A.5 Where the ensemble is heading

A wide trained ensemble converges to the NTK-limit predictive distribution, whose covariance differs from the NNGP posterior covariance:

$$\Sigma_{\text{NTK}} = K_{hh} + A K_{XX} A^T - (A K_{Xh} + K_{hX} A^T), \text{ with } A = \theta_{hX} \theta_{XX}^{-1}$$

Both sides come from initialization alone, so the asymptote is computable at modest width without training anything at scale. Matching the observation-noise ridge across the two expressions:

Width	Posterior, hole	NTK limit, hole	Ratio	Posterior, support	NTK limit, support	Ratio
32	0.0191	0.0113	0.59	0.0104	0.0028	0.27
128	0.0187	0.0110	0.59	0.0104	0.0031	0.30
256	0.0189	0.0109	0.58	0.0104	0.0031	0.30
512	0.0187	0.0108	0.58	0.0103	0.0033	0.32

1024	0.0187	0.0109	0.58	0.0104	0.0034	0.33
------	--------	--------	------	--------	--------	------

The asymptote converges by width 32 and moves at slope -0.013 per doubling thereafter, so its value holds for every larger width. Measured ensemble spread at width 256 remains 8.8x above it, so the descent must pass through the correct posterior on the way down. A crossing exists. Extrapolating the measured ensemble slope places it near width 5600 for this construction, offered as an order of magnitude and nothing more.

Two qualifications matter more than the crossing. The asymptotic overconfidence in the hole is a factor of 1.7, which is a real distortion and not a collapse. And the understatement inside the training support runs worse than outside it, which inverts the expectation that drove the whole construction.

We proposed noise-fitting as the cause of that inversion, tested it, and killed it. Set $\theta = c \cdot K$ in the expression above and it reduces exactly to the NNGP posterior, so the entire gap measures how far the neural tangent kernel departs from proportionality with the NNGP kernel. Measured shape correlation between the two, after unit-diagonal normalization, is 0.9996, with mean absolute residual 0.0354 at width 256 and 0.0137 at width 1024. The residual shrinks with width and the ratio does not, which places the gap in the infinite-width limit rather than in finite-width fluctuation. See T4.

A.6 What the ensemble is actually measuring

Disagreement between trained networks measures how much room the training recipe leaves for two networks to end up different. That is a property of the architecture, the initialization scheme, and the optimizer. It coincides with ignorance only where data removes freedom and ignorance together, which happens inside the support and fails outside it. Outside the support is where the measurement gets used.

Our decomposition locates the width dependence. Holding initialization fixed and varying only batch order gives hole divergence at slope -0.136 , which is flat. Varying initialization and holding batch order fixed gives slope -0.439 , which reproduces the full effect. Initialization carries the collapse and the optimizer contributes a floor near 0.08. Relative kernel drift during training falls from 60.2 at width 8 to 2.2 at width 128, placing the sweep across the feature-learning to lazy transition.

The reachable set narrows with width, the estimator faithfully reports the narrowing, and the narrowing has nothing to do with what the model knows.

A.7 The bridge

The calibration paper asks for the cause of a collapse. The empirical Bayes paper supplies a frame in which the collapse needs no cause: an ensemble estimates a posterior under an implicitly learned prior, that prior belongs to the architecture, and changing the architecture changes the estimate. Our measurement closes the loop by showing that the correctly computed posterior under that same implicit prior stays fixed while the estimate moves. The estimator drifts away from its own target.

For the authors of the calibration paper, this converts an open question into a specification problem. “Why does epistemic uncertainty collapse with complexity” has no answer because the premise is wrong. “How much does ensemble spread deviate from the posterior it approximates, as a function of architecture and region” has an answer and a method.

For the authors of the empirical Bayes paper, this is a consequence of their theorem they did not draw. If the prior is learned rather than chosen, then every architectural knob is a prior knob, and any uncertainty number produced this way inherits the knob. Their result predicts the calibration paper’s paradox.

A third strand supports the same reading from clinical practice. Work in PLOS Digital Health (2025) found that implicit functional prior methods yield systematically biased epistemic uncertainty, and named the difficulty as enforcing functional priors in data-sparse regions without explicit mechanisms. The hole in our construction is a data-sparse region built on purpose, and the bias they report is the 49x we measured.

Failure Ledger

F1. Prior contraction as the explanation. Claude argued that priors converge with width, so wide seeds share an extrapolation and the reported ignorance falls. The first clause holds. The second does not follow, since the posterior under those same priors stays flat. Repair: the collapse belongs to the estimator failing to sample its own posterior, not to any change in the posterior.

F2. Mismatched ridges. Claude compared a noisy-observation NNGP posterior against a ridgeless NTK expression and obtained an asymptote 60x below the posterior. The gap was the ridge, not the physics. Caught by a conditioning check before publication; $\text{cond}(K_{XX})$ runs at $1.2e15$, so the ridgeless branch was reporting the

floating-point floor. Repair: match the observation-noise ridge across both expressions, scaled by the trace ratio. All A.5 numbers use the matched form.

F3. Overstated severity. Claude described the post-crossing regime as confidently wrong before computing the magnitude, which came in at 1.7x. That is a distortion worth knowing and not an alarm. The alarm was rhetoric ahead of a number.

F4. The wrong mechanism for the in-support gap. Claude proposed noise-fitting and wrote it into the paper as the candidate. The disproof was already sitting in a comment in Claude’s own script, noting that proportional kernels make the two expressions identical, so the gap could not be about noise. The comment preceded the paper text and contradicted it. Repair: read the note before writing the claim.

Test Queue

T1. Source decomposition of the width effect. COMPLETE. Prediction: initialization rather than optimizer noise carries the width dependence. Kill condition: width dependence survives with initialization held fixed. Result: init-only slope -0.439 , order-only slope -0.136 . Did not fire. Passed.

T2. Width invariance of the implied posterior. COMPLETE. Prediction: the Gaussian process posterior in the hole, computed under each width’s own implicit prior, stays flat while ensemble spread collapses. Kill condition: the posterior tracks the collapse. Result: posterior slope $+0.004$ against ensemble slope -0.527 , stable across 400, 1000, and 2000 initialization draws. Did not fire. This is the paper.

T3. The crossing point. COMPLETE. Prediction: a crossing exists and the ensemble becomes overconfident above it. Kill condition: ensemble spread flattens at a floor above the posterior and never crosses. Result: the floor sits 42 percent below the posterior, so a crossing follows. Did not fire. Severity came in lower than predicted, and the in-support inversion was not predicted at all.

T4. The in-support understatement. COMPLETE, and discovered rather than sought. Prediction: the gap tracks observation noise, shrinking the in-support ratio toward 1 as noise falls. Kill condition: the ratio holds near 0.30 across a tenfold noise change. Result: at width 1024 the ratio reads 0.303, 0.33, 0.405, 0.45 across noise variances of 0.00025, 0.0025, 0.025, 0.075. Fired over the first tenfold, and across the full 300x range the ratio moved opposite to the prediction. Noise-fitting is dead. The surviving account is kernel shape mismatch.

T5. Architecture generality. OPEN. Test the same comparison on convolutional and attention architectures, and on real data with held-out regions rather than a synthetic hole. Prediction: posterior slope stays within ± 0.1 while ensemble slope stays below -0.3 . Kill condition: posterior slope exceeds -0.2 on any architecture family, meaning the invariance is an artifact of one dimensional tanh networks.

T6. Classification and mutual information. OPEN. Our measurement is a regression variance; the uncertainty literature more often uses mutual information over softmax outputs. Prediction: the same width invariance holds for the classification analogue. Kill condition: implied posterior mutual information falls with width at a rate within a factor of two of the ensemble’s, meaning the result does not transfer off regression.

T7. Depth dependence of the kernel gap. OPEN. The gap between the NTK-limit variance and the NNGP posterior comes from θ failing to be proportional to K , and that failure is a known function of depth. Prediction: the ratio moves toward 1 as depth falls toward 1 and away from 1 as depth grows. Kill condition: the ratio stays fixed across depths 1 through 6, which would place the cause elsewhere and leave us without a mechanism again.

Limitations, In Full

1. One construction, one dimensional input, one architecture family, six widths, eight to twelve seeds. Everything here is synthetic. T5 exists because of this and we would not defend the result past it.
2. The crossing width near 5600 comes from extrapolating a measured slope, not from training at that size. Only the asymptote is computed exactly. Treat the location as an order of magnitude and the existence of the crossing as the result.
3. The NTK asymptote assumes lazy training, and our own kernel-drift measurements show these networks are not lazy at the widths we trained. A real ensemble may never reach the asymptote we computed.
4. Our width range spans 8 to 1024. The uncertainty literature cares about models many orders larger.
5. We compare an ensemble of trained networks against a Gaussian process posterior. These differ in more than one respect at once, since the ensemble also carries optimizer noise and finite training time. The decomposition in A.6 addresses part of this and does not eliminate it.
6. The claim that no scalar rescaling repairs the distortion rests on two probe regions and one construction.
7. We searched the component claims against prior art with more rigor than we applied to the synthesis.

Assume this bridge may exist somewhere and treat the empirical numbers as the contribution.

What Would Change Our Mind

The implied posterior tracking the ensemble collapse on any architecture family kills the central claim outright, because the collapse would then reflect a real change in implied ignorance rather than estimator drift. This is the most likely way the paper is wrong and it is why T5 leads the open queue.

Ensemble spread flattening at a floor above the posterior rather than below it would leave the estimator conservative at every scale, making the result a curiosity about calibration constants rather than a caution about deployment.

Implied posterior mutual information in classification falling at a rate near the ensemble's would confine the result to regression and remove most of its reach.

A demonstration that our sampled kernel estimate flattens the posterior curve would remove the finding entirely. We tested this directly, sweeping the draw count five-fold with no change in slope, and it survived.

Glossary

Epistemic uncertainty. Uncertainty from missing knowledge, in principle removable with more data. Distinguished from aleatoric uncertainty, which comes from noise in the process and does not go away.

Deep ensemble. Several networks with the same architecture trained from different random initializations. Their disagreement on an input serves as the standard practical estimate of epistemic uncertainty.

Implicit prior. The distribution over functions a randomly initialized architecture represents before it sees data. Nobody writes it down; the architecture and initialization scheme choose it.

NNGP kernel. The covariance of the outputs of randomly initialized networks across pairs of inputs. It defines the implicit prior as a Gaussian process.

NTK, neural tangent kernel. The inner product of parameter gradients across pairs of inputs. It governs how a network's function moves during gradient training, and it is not proportional to the NNGP kernel.

Gaussian process posterior. The exact Bayesian answer once a Gaussian process prior is conditioned on data. Its standard deviation at a point is the correct uncertainty under that prior.

The hole. A region of input space deliberately left with no training data, so that model behavior there reflects inductive bias alone.

Reachable set. The functions a given training recipe can actually produce. Narrower than the set of functions consistent with the data, and its width depends on architecture rather than evidence.

Lazy regime. Training in which the network's effective kernel barely moves from its initialization value. Wide networks approach it; narrow networks do not.

Changelog

Version 1.0, July 26, 2026. First release. Supersedes nothing. Retires an earlier unpublished framework on representation systems under finite description budgets, whose ledger stays with it; the support-hole construction in A.1 is the only technical content carried forward.

Provenance

Sam supplied the framing that produced the result: the proposal that the collapse might be a property of the act of measuring rather than of the thing measured, stated before any Gaussian process test existed. Sam also directed the search discipline that retired the earlier framework, and set the editorial decision to publish as a bridge rather than as a claim.

Claude supplied the constructions, the code, the decomposition design, and the Gaussian process and NTK comparisons. Claude also supplied all four entries in the failure ledger.

Borrowed results, none of which we claim. The empirical Bayes reading of deep ensembles comes from arXiv:2501.17917. The collapse phenomenon and its open question come from arXiv:2407.12211 and from *Epistemic Uncertainty Collapse in Large Models* (OpenReview hR3mhl2p2K). The decomposition of ensemble uncertainty into data variability and training stochasticity is formalized in arXiv:2510.22063. The clinical evidence

on implicit functional priors in data-sparse regions comes from PLOS Digital Health (2025). Kernel concentration at $1/\sqrt{w}$, the lazy training regime, and the NTK-limit predictive distribution come from the neural tangent kernel literature.

Data and reproduction. All numbers come from CPU runs in a single container on July 26, 2026, using PyTorch 2.13. The construction, widths, seed counts, and probe regions appear in A.1. The Gaussian process comparison uses 400, 1000, and 2000 initialization draws per width with matching results, 160 conditioning points, and 40 probe points per region. The NTK computation in A.5 uses exact per-point parameter gradients at initialization, averaged over 3 seeds, on 100 conditioning points and 16 probe points per region. Scripts and raw logs accompany this paper.

A measurement that reports faithfully on the wrong quantity is harder to catch than one that reports badly on the right one.

License: CC BY 4.0. Knowledge should never die unpublished.